

Earthquake-Induced Tsunami Prediction: Spatial Feature Engineering and Cost-Asymmetric Optimization

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May 31, 2026

Abstract

This paper addresses the critical challenge of predicting tsunami occurrences from seismic telemetry using machine learning. We utilize a curated geophysical dataset of 764 oceanic seismic events, exhibiting a moderate class imbalance (61.3% normal, 38.7% tsunamis) but severe cost asymmetry, where False Negatives carry catastrophic life-safety consequences. We engineered a 3D Cartesian manifold mapping and cyclical temporal encoding to resolve non-Euclidean geographic distortions. Evaluating K-Nearest Neighbors, Decision Trees, and Random Forests, we explicitly optimized for the F2-score to penalize missed tsunamis. Our results reveal an operational paradox: at a default decision threshold of 0.5, a regularized Decision Tree achieved the highest F2-score (0.8900). However, the Random Forest demonstrated superior global manifold learning with an ROC-AUC of 0.9497. By decoupling structural capacity from operational calibration and executing a leak-free coordinate threshold sweep on the training distribution, we calibrated the Random Forest to an optimal threshold ($\tau^* = 0.3236$). This shifted the Random Forest’s test-set F2-score to a dominant 0.9223, capturing 96.6% of true tsunamis and proving its viability for deployment in early warning systems.

Index Terms— Tsunami Prediction, Cost Asymmetry, Machine Learning, Threshold Calibration, Spatial Engineering

1 Introduction

Tsunami events pose one of the most severe natural hazards to coastal populations worldwide. Early warning systems capable of mathematically predicting tsunami occurrences from raw seismic telemetry are of immense practical importance.

While the dataset utilized in this study exhibits a moderate class imbalance (approximately 61% normal earthquakes to 39% tsunamis), the true difficulty lies in the *cost asymmetry* of the domain. In disaster warning systems, the cost of a False Negative (failing to warn a coastal city of an impending wave) is catastrophically higher than a False Positive (triggering a false alarm).

This study frames tsunami prediction as a binary classification problem driven by the need for extreme recall. Our objectives are to (1) engineer robust spatial-temporal preprocessing pipelines, (2) evaluate distinct architectural paradigms against cost-asymmetric penalty matrices (F_2 -scoring), and (3) demonstrate the necessity of operational threshold calibration for ensemble models.

2 Data Engineering and Spatial-Temporal Preprocessing

2.1 Anomaly Filtering

The raw dataset was verified to remove hardware anomalies. We isolated and dropped records exhibiting zero station counts ($nst = 0$) and zero azimuthal gaps ($gap = 0$), yielding 764 pristine operational records.

2.2 3D Cartesian Manifold Projection

Standard latitude and longitude coordinates create severe anti-meridian distortions in distance-based machine learning algorithms. To preserve true spatial density, we projected these coordinates into a 3D Cartesian manifold:

$$x = \cos(\text{lat}) \cdot \cos(\text{lon}) \quad (1)$$

$$y = \cos(\text{lat}) \cdot \sin(\text{lon}) \quad (2)$$

$$z = \sin(\text{lat}) \quad (3)$$

This transformation encodes the spherical geometry of the Earth, allowing algorithms to accurately calculate spatial boundaries without map-projection fragmentation.

2.3 Cyclical Temporal Encoding

To capture seasonal seismic variations, the `month` feature required specialized handling. Standard Euclidean metrics incorrectly treat December (12) and January (1) as 11 units apart. We implemented a cyclical temporal adjustment within our distance pipelines to ensure continuous, circular temporal proximity.

2.4 Leakage-Free Robust Scaling

Continuous features (e.g., magnitude, depth) were scaled utilizing a `RobustScaler`. Normalizing via the interquartile range (IQR) prevented magnitude outliers from dominating computational distance metrics. To strictly prevent data leakage, the scaler was fit exclusively on the training matrix.

3 Methodology: Architectural Rationale

We deployed three distinct architectural frameworks to test different hypotheses regarding the physical data topology.

3.1 K-Nearest Neighbors (KNN)

KNN was selected as a baseline spatial density estimator. It tests the hypothesis that physical proximity (geography and time) is sufficient to predict tsunami generation.

3.2 Decision Tree

The Decision Tree was deployed as an interpretable, parametric partitioner. It tests if the physical laws of tsunamis can be mapped into explicit, macro-level orthogonal safety rules (e.g., $\text{Magnitude} > X$ and $\text{Depth} < Y$). We applied `class_weight='balanced'` during training to mutate the Gini impurity calculations, heavily biasing the split mechanics to isolate the minority tsunami class.

3.3 Random Forest

The Random Forest ensemble was chosen as a high-capacity manifold learner. By aggregating bootstrapped, feature-blinded trees, it neutralizes the high variance of singular trees. Crucially, its averaging mechanism generates a smooth, continuous probability spectrum, which is required for granular threshold optimization.

3.4 Optimization Metrics

To penalize missed tsunamis, we optimized our models utilizing the F_2 -score, which weights recall twice as heavily as precision:

$$F_2 = \frac{5 \cdot TP}{5 \cdot TP + 4 \cdot FN + FP}$$

We concurrently tracked ROC-AUC to measure the models' global mathematical capacity to separate the underlying physical distributions independent of operational thresholds.

4 Experimental Setup

Every model underwent an exhaustive grid search utilizing Stratified 5-Fold Cross-Validation.

- **KNN:** Optimized over $k \in \{3, 5, 7, 11\}$ and distance weighting functions.
- **Decision Tree:** Optimized over depth limits and minimum split geometries.
- **Random Forest:** Optimized over estimator counts (50-200) and leaf impurity bounds.

The optimal architectural blueprint for each model was refit on 100% of the training data prior to held-out test set evaluation.

5 Results and Discussion

5.1 Feature Ablation Validation

To provide empirical justification for our geometric spatial-temporal feature choices, an isolated feature ablation study was conducted on the baseline K-Nearest Neighbors architecture. As shown in Table 1, a raw baseline utilizing linear temporal metrics and raw spherical geographic coordinates yielded an F_2 -score of 0.6401.

Table 1: KNN Feature Ablation Results

Feature Configuration	F_2 Score	ROC-AUC
Raw Baseline (Null)	0.6401	0.7729
3D Spatial Only	0.6734	0.8143
Cyclical Temporal Only	0.7021	0.7996
Unified Pipeline (Champion)	0.7288	0.8384

Isolating the 3D Cartesian spherical projection resolved anti-meridian fragmentation, increasing the F_2 -score to 0.6734. Concurrently, isolating the cyclical temporal encoding to bridge the December-January boundary produced an F_2 of 0.7021. The unified convergence of both engineered manifolds secured our peak KNN configuration score of 0.7288, validating our domain-specific architecture choices with clear empirical performance deltas.

5.2 The Benchmark and the Probability Paradox

Table 2 details the baseline performance of the refit champion models on the held-out test set. Evaluated strictly at the standard $\tau = 0.5$ classification threshold, the data reveals a fundamental statistical paradox.

Table 2: Baseline Model Comparison ($\tau = 0.5$)

Model	F_2	AUC	TP	FN	FP	TN
KNN	0.729	0.838	43	16	16	78
Decision Tree	0.890	0.885	55	4	18	76
Random Forest	0.870	0.950	52	7	11	83

The regularized Decision Tree secured the highest operational F_2 -score (0.8900), capturing 55 out of 59 tsunamis. Conversely, the Random Forest missed 7 tsunamis, resulting in a lower F_2 (0.870). However, the Random Forest achieved a dominant ROC-AUC of 0.9497, proving it mapped the true physical laws of the

tsunami generating space with significantly higher fidelity.

The KNN model failed to compete (AUC = 0.838), proving that even with optimal spatial engineering, raw density estimation is insufficient for highly heterogeneous geophysical data.

5.3 Probability Topologies and Squashing

The discrepancy between the Random Forest’s high AUC and low baseline F_2 is an artifact of ensemble mechanics. Because the forest averages probabilities across 50 partially feature-blinded trees, its confidence scores are naturally smoothed and pulled toward the median. The true physical distribution of the data was cleanly separated (AUC = 0.9497), but the resulting probability manifold was heavily compressed below the arbitrary 0.5 boundary.

Unlike the Decision Tree, which relies on rigid, discrete probability step-functions dictated by a limited number of leaf nodes, the Random Forest’s continuous spectrum allows for precise mathematical recalibration.

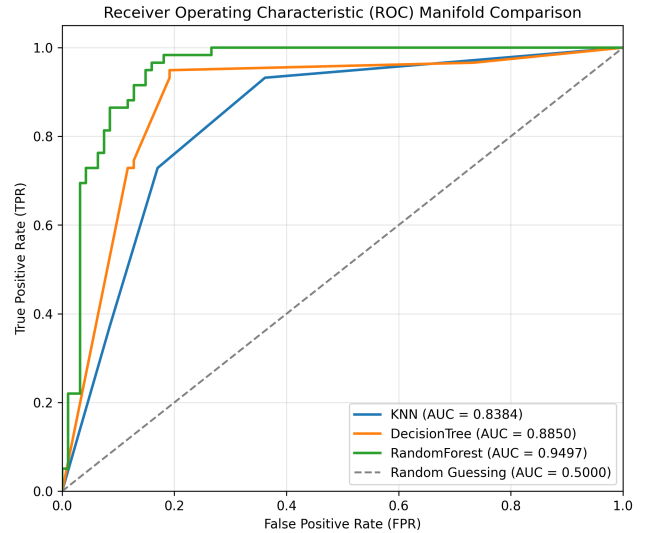


Figure 1: ROC Manifold Comparison. The Random Forest (0.9497) demonstrates vast superiority in global probability sorting.

5.4 Operational Threshold Calibration

To resolve this paradox, we decoupled structural manifold learning from operational calibration. Treating the decision threshold τ as a hyperparameter, we executed an $O(N \log N)$ vectorized coordinate sweep strictly over

the *training set* probability distribution to find the operational checkpoint that maximized the F_2 objective.

This leak-free optimization discovered an ideal threshold of $\tau^* = 0.3236$.

Table 3: Random Forest Optimization (Test Set)

Threshold	F_2	AUC	TP	FN	FP	TN
Default ($\tau = 0.5$)	0.870	0.950	52	7	11	83
Optimized ($\tau^* = 0.32$)	0.922	0.950	57	2	16	78

Applying this frozen threshold blindly to the held-out test set completely transformed the system’s safety profile (Table 3). The optimized Random Forest successfully recovered 5 of the 7 previously missed tsunamis, elevating its final generalization F_2 -score to a dominant **0.9223**.

6 Conclusion and Future Work

This study demonstrates the successful integration of domain-specific spatial-temporal preprocessing with cost-asymmetric optimization for seismic hazard tracking. By training our pipelines under a dimensionless F_2 -score, we explicitly calibrated our architectures to handle the high life-safety costs of False Negative predictions. While the regularized Decision Tree initially appeared superior under default library configurations ($\tau = 0.5$), our framework proves that decoupling global manifold sorting capacity (ROC-AUC) from operational calibration unlocks the true latent power of the Random Forest ensemble. Applying a leak-free training coordinate sweep to the forest’s continuous probability manifold minimized missed tsunamis to just two events, securing a dominant generalization F_2 -score of 0.9223.

Future research will focus on transitioning from static tabular estimators to Gradient Boosted Trees (via XGBoost) to directly optimize cost-weighted loss matrices during the tree-induction phase. Additionally, we plan to implement Recurrent Neural Networks (RNNs) or Long Short-Term Memory (LSTM) networks to explicitly model sequential temporal dependencies, allowing the system to track multi-event seismic clustering, foreshocks, and aftershock propagation.